

JavaScript Core Concepts – Structured Notes

ARRAY METHODS (Used for Arrays)

- `map()` → Transforms every element and returns a new array (same length).
- `filter()` → Returns a new array with elements that match a condition.
- `reduce()` → Combines all elements into one single value.
- `forEach()` → Executes a function for each element (returns nothing).
- `find()` → Returns the first matching element.
- `includes()` → Checks if value exists (true/false).
- `push()` → Adds element to end.
- `pop()` → Removes last element.
- `shift()` → Removes first element.
- `unshift()` → Adds element to beginning.
- `slice()` → Returns portion of array without modifying original.
- `splice()` → Adds/removes elements inside array.
- `sort()` → Sorts array elements.
- `join()` → Converts array into string.

STRING METHODS (Used for Strings)

- `toUpperCase()` → Converts string to uppercase.
- `toLowerCase()` → Converts string to lowercase.
- `trim()` → Removes extra spaces from start and end.
- `slice()` → Extracts part of string.
- `split()` → Converts string into array.
- `replace()` → Replaces part of string.
- `includes()` → Checks if text exists in string.
- `concat()` → Joins strings together.

LOOPS (Used for Repeating Tasks)

- `for` loop → Used when number of iterations is known.
- `while` loop → Runs while condition is true.
- `do...while` loop → Runs at least once before checking condition.
- `forEach()` → Used to iterate over arrays only.
- `for...of` → Used to iterate over array values.
- `for...in` → Used to iterate over object keys.

OBJECTS (Used for Key-Value Data)

- `Object.keys()` → Returns array of object keys.
- `Object.values()` → Returns array of object values.
- `Object.entries()` → Returns array of key-value pairs.
- `Destructuring` → Extract values from objects easily.
- `Spread operator (...)` → Copies or merges objects.

FUNCTION TYPES

- Function Declaration → Traditional named function.
- Function Expression → Function stored in variable.
- Arrow Function → Short syntax function.
- Callback Function → Function passed as argument.
- Higher-Order Function → Function that takes/returns another function.

IMPORTANT CORE CONCEPTS

- Variables (let, const, var) → Store data values.
- Data Types → String, Number, Boolean, Object, Array, Null, Undefined.
- Operators → Arithmetic, Comparison, Logical operators.
- Conditionals → if, else, switch for decision making.
- DOM Manipulation → Interact with HTML using JavaScript.
- Events → Handle user actions (click, input, submit).
- Promises → Handle asynchronous operations.
- Async/Await → Cleaner syntax for async code.